Assignment 3

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Ques1) Only sequential

```
saksham@saksham:~/Desktop/parallel-assignments/Lab3$ ./correlate 500 2000 1 seq
==============================================
  Pairwise Correlation Assignment
==============================================
Matrix size : 500 x 2000
OMP threads : 1
Mode        : seq
Result size : 500 x 500 (upper triangle)

--- Part 1: Sequential ---
Time: 138.219 ms
```

Ques2) Parallely running on 4 threads

```
saksham@saksham:~/Desktop/parallel-assignments/Lab3$ ./correlate 500 2000 4 par
==============================================
  Pairwise Correlation Assignment
==============================================
Matrix size : 500 x 2000
OMP threads : 4
Mode        : par
Result size : 500 x 500 (upper triangle)


--- Part 2: OpenMP Parallel ---
Time: 40.717 ms
```

Ques3) Running as fast as possible

```
saksham@saksham:~/Desktop/parallel-assignments/Lab3$ ./correlate 500 2000 4 opt
==============================================
  Pairwise Correlation Assignment
==============================================
Matrix size : 500 x 2000
OMP threads : 4
Mode        : opt
Result size : 500 x 500 (upper triangle)


--- Part 3: Optimized (SIMD + OpenMP + cache) ---
Time: 41.776 ms
```

Ques4)

Sequential

```
saksham@saksham:~/Desktop/parallel-assignments/Lab3$ perf stat ./correlate 500 2000 1
============================================
  Pairwise Correlation Assignment
============================================
Matrix size : 500 x 2000
OMP threads : 1
Mode        : all
Result size : 500 x 500 (upper triangle)

--- Part 1: Sequential ---
Time: 137.001 ms

--- Part 2: OpenMP Parallel ---
Time: 133.94 ms
Speedup vs Sequential: 1.02285x
Correctness: PASS

--- Part 3: Optimized (SIMD + OpenMP + cache) ---
Time: 136.519 ms
Speedup vs Sequential: 1.00353x
Correctness: PASS

============================================
  Summary
============================================
Sequential : 137.001 ms
Parallel   : 133.94 ms  (speedup 1.02285x)
Optimized  : 136.519 ms  (speedup 1.00353x)
============================================

 Performance counter stats for './correlate 500 2000 1':

      42,54,98,567      task-clock                       #    0.995 CPUs utilized
                14      context-switches                 #   32.903 /sec
                 1      cpu-migrations                   #    2.350 /sec
             7,703      page-faults                      #   18.103 K/sec
      44,33,33,436      cpu_atom/instructions/           #    1.15  insn per cycle              (0.62%)
    3,17,48,92,459      cpu_core/instructions/           #    2.03  insn per cycle              (99.15%)
      38,55,80,163      cpu_atom/cycles/                 #    0.906 GHz                         (0.62%)
    1,56,30,02,585      cpu_core/cycles/                 #    3.673 GHz                         (99.15%)
       8,28,53,313      cpu_atom/branches/               #  194.721 M/sec                       (0.61%)
      40,17,58,726      cpu_core/branches/               #  944.207 M/sec                       (99.15%)
          23,77,663      cpu_atom/branch-misses/          #    2.87% of all branches             (0.81%)
           4,03,501      cpu_core/branch-misses/          #    0.10% of all branches             (99.15%)
 #     56.6 %  tma_backend_bound
                                                     #      0.4 %  tma_bad_speculation
                                                     #      7.8 %  tma_frontend_bound
                                                     #     35.2 %  tma_retiring             (99.15%)
 #     17.9 %  tma_bad_speculation
                                                     #     26.6 %  tma_retiring             (0.85%)
 #     29.6 %  tma_backend_bound
                                                     #     25.9 %  tma_frontend_bound       (0.85%)

       0.427476569 seconds time elapsed

       0.410017000 seconds user
       0.017000000 seconds sys
```

Parallel – 4 threads

```
saksham@saksham:~/Desktop/parallel-assignments/Lab3$ perf stat ./correlate 500 2000 4
==============================================
  Pairwise Correlation Assignment
==============================================
Matrix size : 500 x 2000
OMP threads : 4
Mode        : all
Result size : 500 x 500 (upper triangle)

--- Part 1: Sequential ---
Time: 137.417 ms

--- Part 2: OpenMP Parallel ---
Time: 48.92 ms
Speedup vs Sequential: 2.80901x
Correctness: PASS

--- Part 3: Optimized (SIMD + OpenMP + cache) ---
Time: 43.251 ms
Speedup vs Sequential: 3.1772x
Correctness: PASS

==============================================
  Summary
==============================================
Sequential : 137.417 ms
Parallel   : 48.92 ms  (speedup 2.80901x)
Optimized  : 43.251 ms  (speedup 3.1772x)
==============================================

 Performance counter stats for './correlate 500 2000 4':

      49,33,67,028      task-clock                       #    2.047 CPUs utilized
               39       context-switches                 #   79.049 /sec
                9       cpu-migrations                   #   18.242 /sec
            7,707       page-faults                      #   15.621 K/sec
    1,71,58,02,791      cpu_atom/instructions/           #    1.32  insn per cycle              (23.48%)
    3,68,49,60,386      cpu_core/instructions/           #    2.03  insn per cycle              (72.89%)
    1,29,82,04,107      cpu_atom/cycles/                 #    2.631 GHz                         (23.47%)
    1,81,75,60,944      cpu_core/cycles/                 #    3.684 GHz                         (72.89%)
      21,40,77,305      cpu_atom/branches/               #  433.911 M/sec                       (23.46%)
      46,74,62,189      cpu_core/branches/               #  947.494 M/sec                       (72.89%)
          2,48,016      cpu_atom/branch-misses/          #    0.12% of all branches             (23.84%)
          4,98,240      cpu_core/branch-misses/          #    0.11% of all branches             (72.89%)
 #     56.2 %  tma_backend_bound
                                                     #      0.7 %  tma_bad_speculation
                                                     #      7.9 %  tma_frontend_bound
                                                     #     35.2 %  tma_retiring             (72.89%)
 #      0.4 %  tma_bad_speculation
                                                     #     28.9 %  tma_retiring             (23.88%)
 #     69.3 %  tma_backend_bound
                                                     #      1.5 %  tma_frontend_bound       (23.90%)

       0.241058557 seconds time elapsed

       0.481694000 seconds user
       0.014049000 seconds sys
```

Observations: We can see from the perf stat metrics that the no of CPUs utilized are more (~2 compared to ~1) and the context-switches per sec is also significantly larger. Speedup of 2.8x is achieved through 4 threads.

Result: Execution time increases steeply with matrix size, consistent with the quadratic growth in pairwise comparisons. Small matrices are dominated by thread and memory overhead, while larger matrices expose the memory-bandwidth bottleneck of the kernel. The results confirm that the implementation scales correctly and behaves as expected for a high-performance numerical workload.